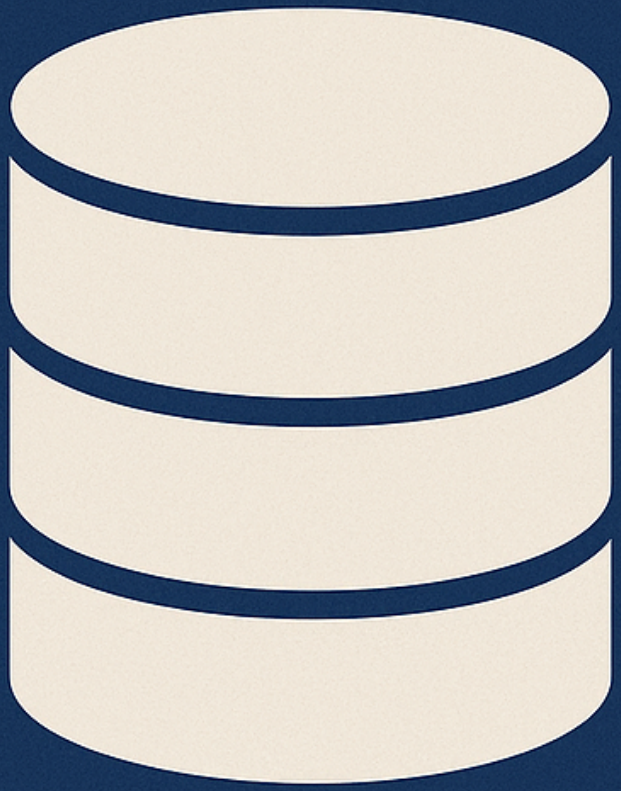
SQL

Class starts
@ 9:10pm





Lecture 2: Data Extraction with SQL

#Agenda

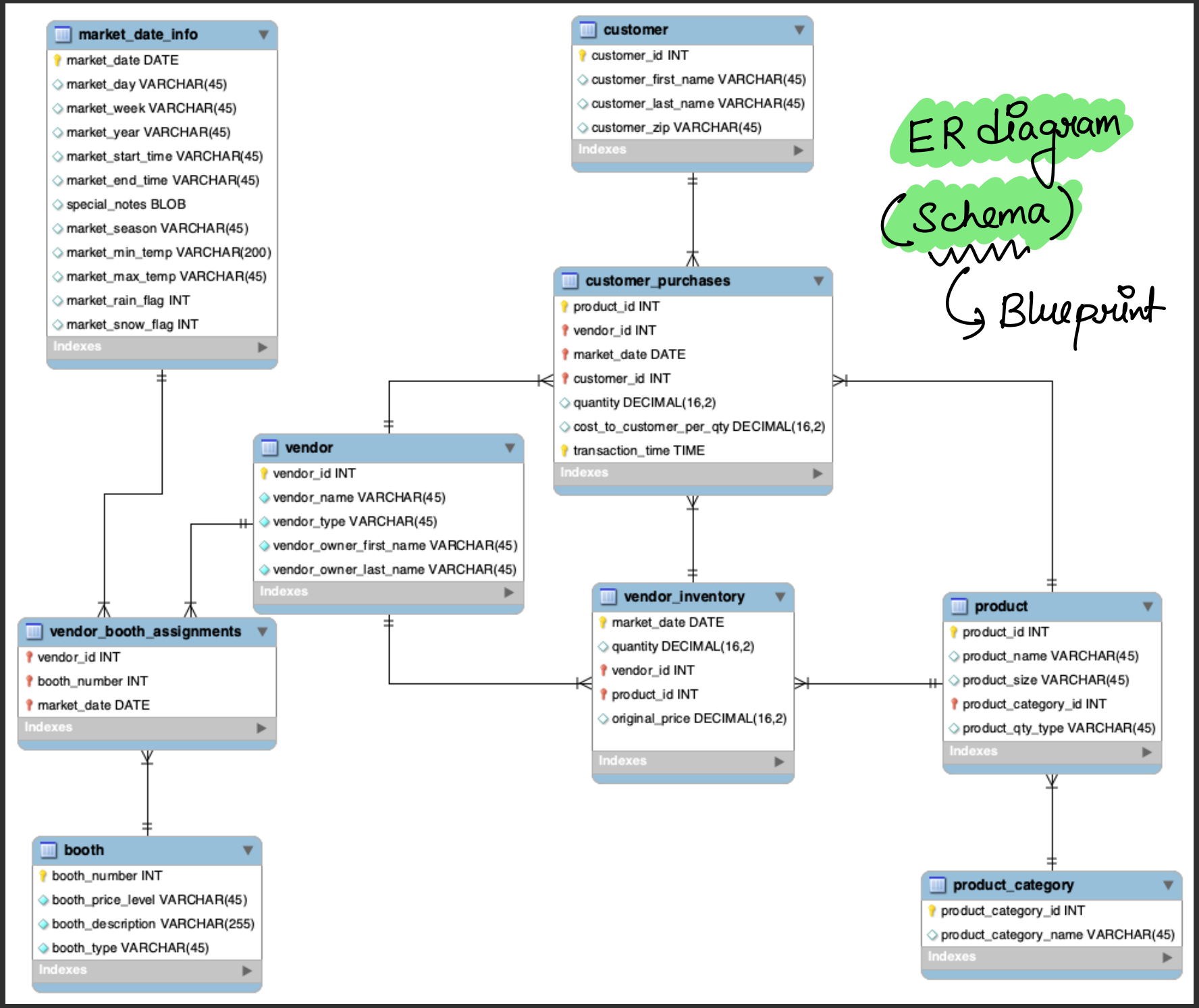
1) ER diagram (ER → Entity Relationship)
2) Relationships
3) Select Statement
4) Order By
5) limit / OFFSET
6) Alias
7) Other fun^{n}

Amazon Farmers Market (AFM), a beloved community market, faces a growing threat from large grocery chains that are expanding their local produce offerings.

It’s your first day as a Data Analyst Intern at Amazon Fresh.

Amazon Fresh has recently decided to open big stores where farmers can directly sell their produce from their stores.

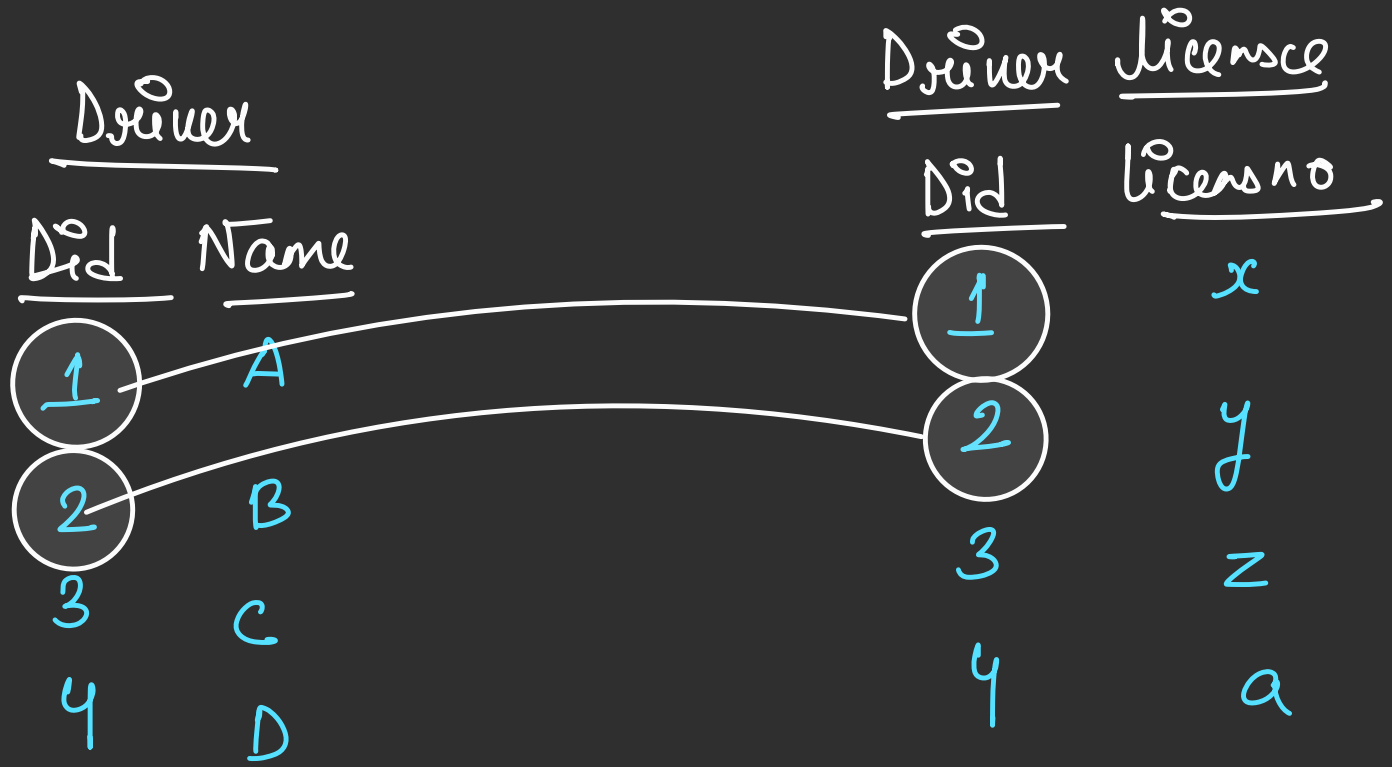
All the data is stored in a Database.


market_date_info
market_date DATE
market_day VARCHAR(45)
market_week VARCHAR(45)
market_year VARCHAR(45)
market_start_time VARCHAR(45)
market_end_time VARCHAR(45)
special_notes BLOB
market_season VARCHAR(45)
market_min_temp VARCHAR(200)
market_max_temp VARCHAR(45)
market_rain_flag INT
market_snow_flag INT
Indexes
customer
customer_id INT
customer_first_name VARCHAR(45)
customer_last_name VARCHAR(45)
customer_zip VARCHAR(45)
Indexes
ER diagram
(Schema)
Blueprint
customer_purchases
product_id INT
vendor_id INT
market_date DATE
customer_id INT
quantity DECIMAL(16,2)
cost_to_customer_per_qty DECIMAL(16,2)
transaction_time TIME
Indexes
vendor
vendor_id INT
vendor_name VARCHAR(45)
vendor_type VARCHAR(45)
vendor_owner_first_name VARCHAR(45)
vendor_owner_last_name VARCHAR(45)
Indexes
vendor_booth_assignments
vendor_id INT
booth_number INT
market_date DATE
Indexes
vendor_inventory
market_date DATE
quantity DECIMAL(16,2)
vendor_id INT
product_id INT
original_price DECIMAL(16,2)
Indexes
product
product_id INT
product_name VARCHAR(45)
product_size VARCHAR(45)
product_category_id INT
product_qty_type VARCHAR(45)
Indexes
booth
booth_number INT
booth_price_level VARCHAR(45)
booth_description VARCHAR(255)
booth_type VARCHAR(45)
Indexes
product_category
product_category_id INT
product_category_name VARCHAR(45)
Indexes

Types of Relationships

(Not used much)

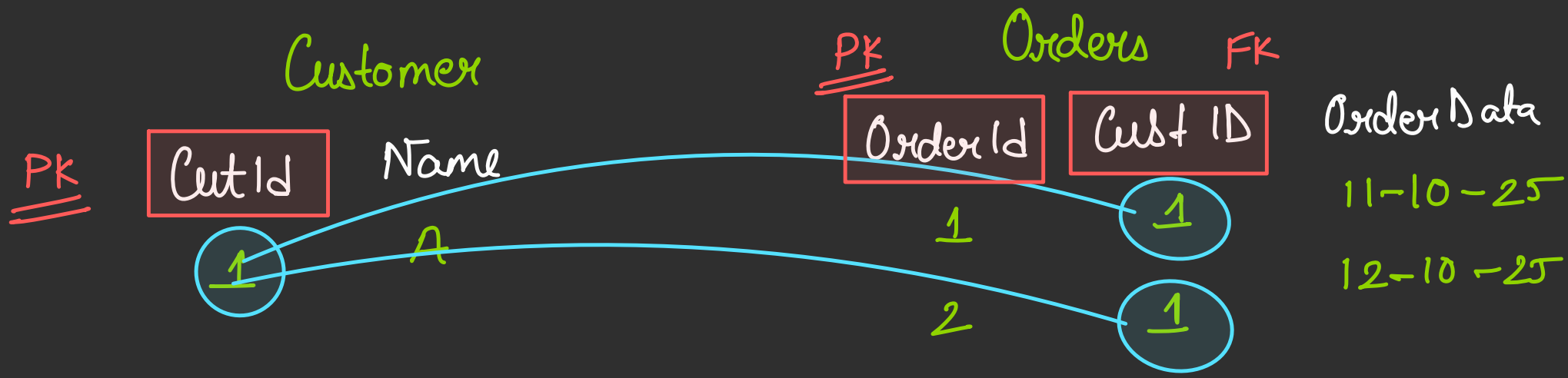
1) One to One Relationship



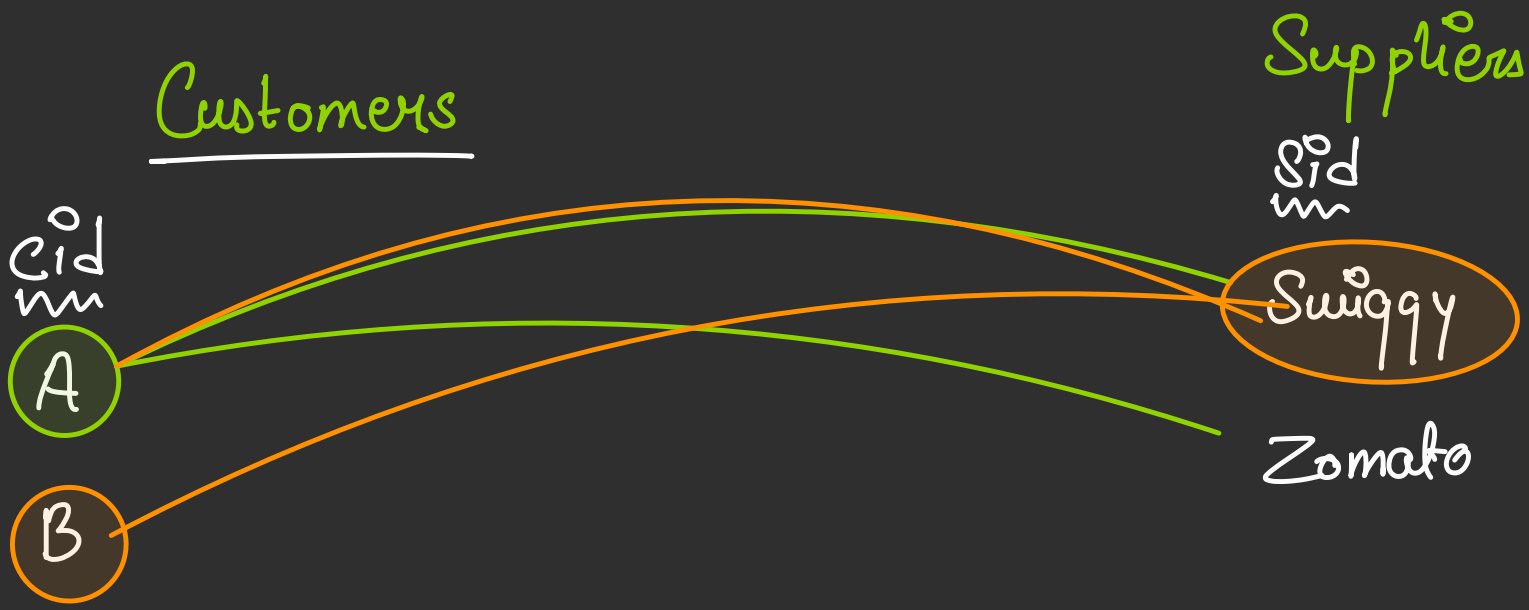
Driver

Did	Name	License
1	A	x
2	B	y
3	C	z
4	D	a

2) One to Many Relationship

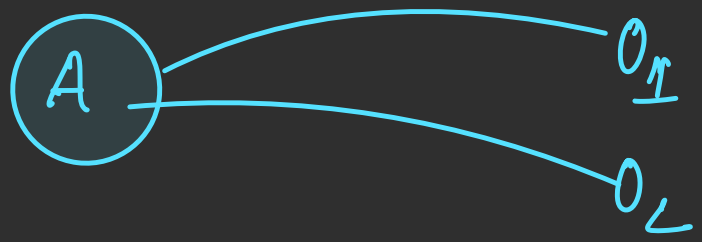


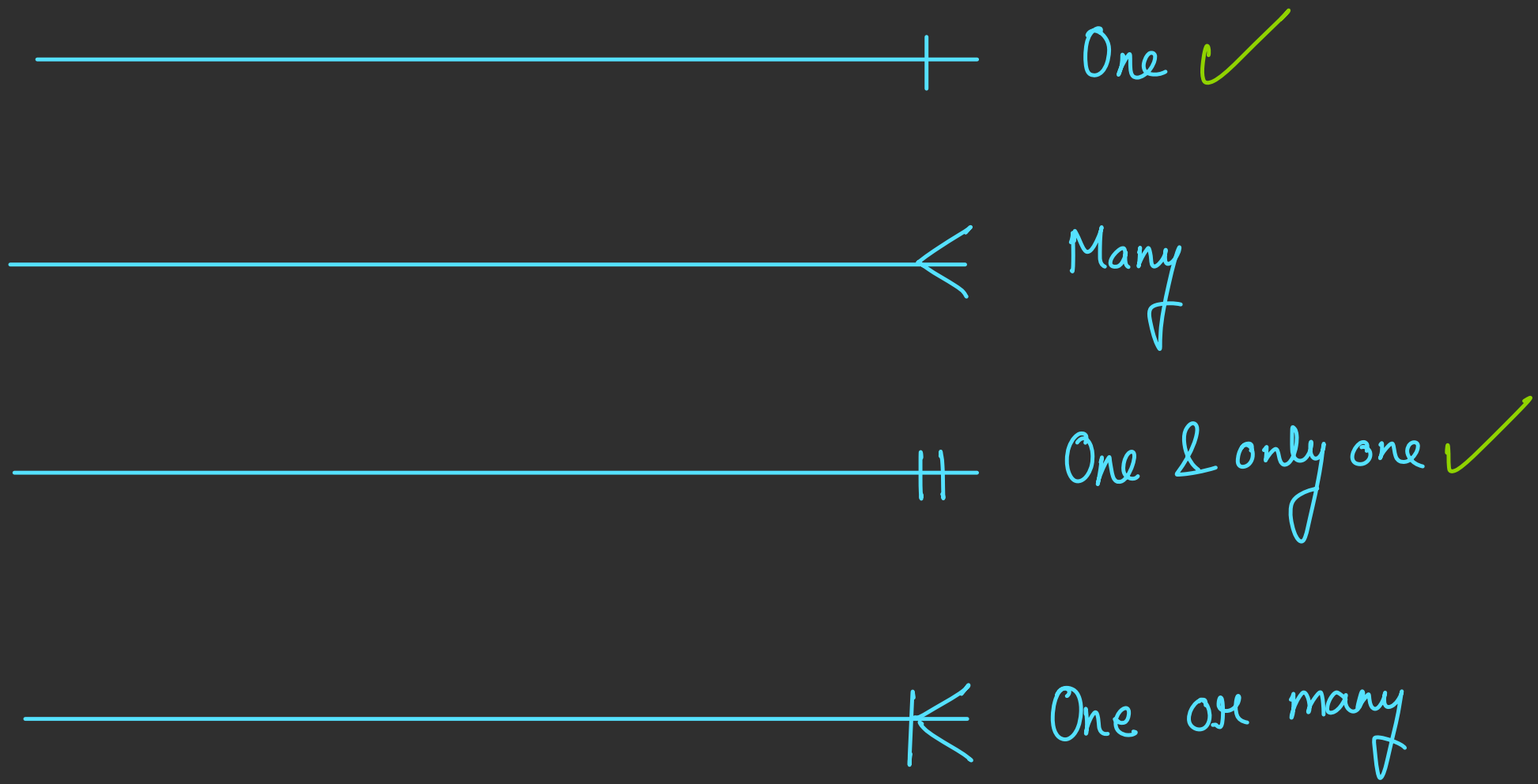
3) Many to Many Relationship



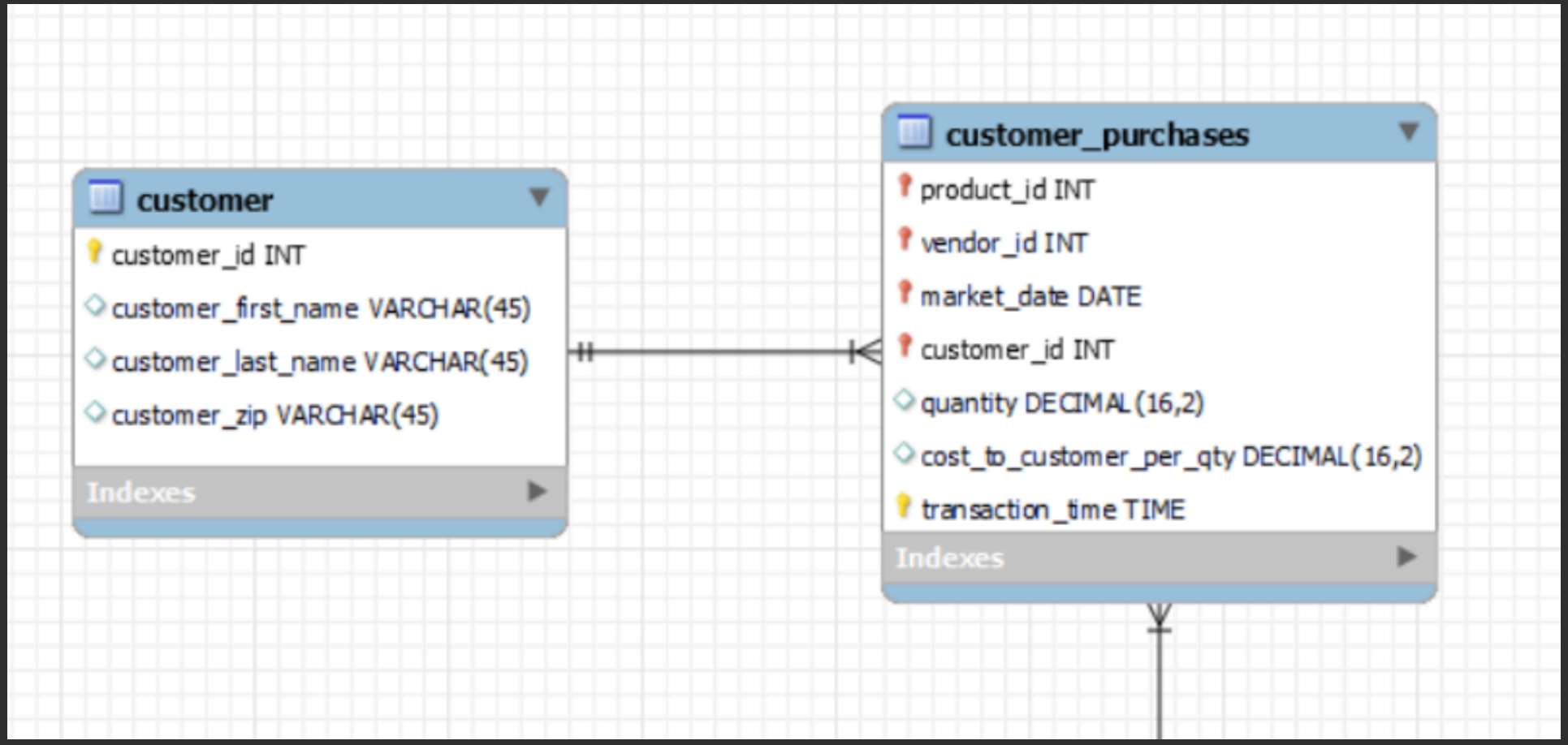
customers & rapido captains

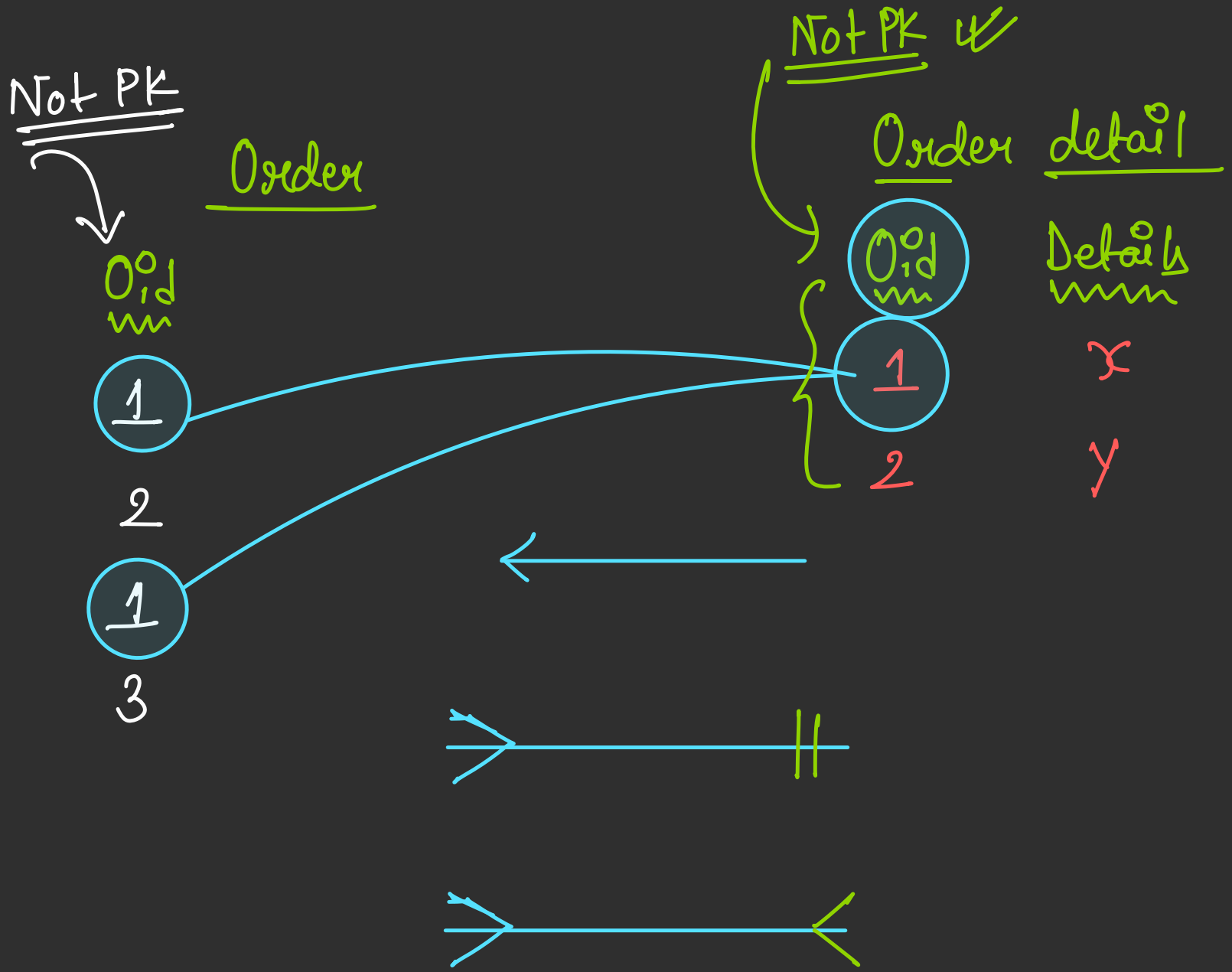
Customer & Order




One
Many
One & only one
One or many

ER Diagram (Entity-Relationship) {ERD}




Not PK
Order
Oid
1
2
1
3
Not PK
Order detail
Oid
Details
1
2
x
y

Customer

Order

"Easy to maintain"

Data Types

Tables ⟶ row & column

Email, Age, Name, Joining date, Salary

(text) (Num) (text) (date) (Decimal) → Normal

(string) (integer) (date) (float) → program ways

13

13.62
13.0

primary key

customer_id	name	phone	age
1	Akon	9876723452	17
2	Akon	9991165674	19
3	Bkon	7898756543	18
4	Ckon	8987867898	19
5	Dkon	9990080080	//////

"Only 1 primary key per table".

Null value
(Absence of data)

Name = ? X

phone Number = X

Age = X

customer_id

Unique & Not Null

primary key

PK

Foreign key = Can have duplicates + Can have Null valu

order_id	item	quantity	cust_id
1	Fruits	2.5	1
2	Veggies	3	1
3	Meat	1	3
4	Veggies	5	5
5	Dairy	2	2

orders

PK

customer_id	name	phone	age
1	Akon	9876723452	17
2	Akon	9991165674	19
3	Bkon	7898756543	18
4	Ckon	8987867898	19
5	Dkon	9990080080	

customer

SQL

(Tabular Data : Rows & Columns)

Select clause (columns)

```
SELECT
[Columns to displays]
```

Modifies the no of rows

```
FROM
"projet_id.dataset_id.table_id"
```

product id	vender id
1	6
2	7
1	9
3	4

Order by product

productid	vendorid
1	6
1	9
2	7
3	4

Order by vendor id desc

productid	uid
1	9
1	6
2	7
3	4